

MEDICAID AND MORTALITY: NEW EVIDENCE FROM LINKED SURVEY AND ADMINISTRATIVE DATA*

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Abstract

We use large-scale federal survey data linked to administrative death records to investigate the relationship between Medicaid enrollment and mortality. Our analysis compares changes in mortality for near-elderly adults in states with and without Affordable Care Act Medicaid expansions. We identify adults most likely to benefit using survey information on socioeconomic status, citizenship status, and public program participation. We find that, prior to the ACA expansions, mortality rates across expansion and non-expansion states trended similarly, but beginning in the first year of the policy, there were significant reductions in mortality in states that opted to expand relative to non-expanders. Individuals in expansion states experienced a 0.132 percentage point decline in annual mortality, a 9.4 percent reduction over the sample mean, as a result of the Medicaid expansions. The effect is driven by a reduction in disease-related deaths and grows over time. A variety of alternative specifications, methods of inference, placebo tests, and sample definitions confirm our main result. *JEL Codes:* I13, I18, H75.

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I Introduction

Low-income individuals in the U.S. experience dramatically worse health than those with high incomes. For example, the annual mortality rate for older adults in families earning less than 138 percent of the Federal Poverty Level (FPL) is 1.7 percent, more than 4 times higher than the 0.4 percent rate experienced by higher-income individuals of the same age.¹ This low-income group also experiences higher risks of dying from diabetes (by 787%), cardiovascular disease (552%), and respiratory disease (813%) in a given year relative to those in higher income families, despite the existence of effective medical treatment and management for these types of chronic conditions. These higher rates of death translate to dramatic differences in life expectancy across income groups. [Chetty et al. \(2016\)](#) find that men at the bottom of the income distribution live on average nearly 15 years less, and women over 10 years less, than those at the top of the income distribution conditional on surviving to age 40. While data from nearly all countries show a positive correlation between income and health, this correlation is stronger in the U.S. than other high income countries ([Semyonov et al., 2013](#)).

One policy that could play an important role in reducing these disparities is the Medicaid eligibility expansion of the Affordable Care Act (ACA). The ACA brought about the largest expansion of health insurance coverage in the U.S. since the creation of Medicare and Medicaid in 1965. Over 20 million people have gained coverage since the ACA ([Martinez et al., 2018](#)), the majority of whom are low-income adults receiving coverage through the Medicaid program ([Frean et al., 2017](#)). Other research has shown that this expansion in coverage improved access to effective medical care that beneficiaries otherwise would not receive, including mortality-reducing prescription drugs ([Ghosh et al., 2019](#)), earlier detection and treatment of treatable cancers (e.g. [Eguia et al., 2018](#); [Sabik et al., 2018](#); [Soni et al., 2018](#)), and hospital and emergency department visits for conditions that require immediate care ([Duggan et al., 2019](#)), in addition to a variety of other types of potentially beneficial medical care (see [Gruber and Sommers, 2019](#)). However, despite the size and scope of this policy, its impact on the health of those who benefited remains uncertain.

A major challenge to understanding how the ACA affects health is the limited availability of large-scale data that links individual-level information on eligibility for expanded coverage to objective measures of health. Analyses of survey data can use information on individual characteristics to examine eligible adults, but such data are typically limited to self-reported health measures, which may not correlate with actual changes in health. Studies using these measures have offered inconsistent findings, with some indicating that the ACA expansions improved self-reported health, while others found no effect or even negative impacts of the ACA expansions (e.g. [Cawley et al., 2018](#); [Courtemanche et al., 2018b](#); [Miller and Wherry, 2017](#); [Sommers et al., 2017](#)). Meanwhile, analyses of mortality data rely on death records aggregated to the state- or county-level, without information on individual-level factors that determine eligibility such as income (e.g. [Borgschulte and Vogler, 2020](#); [Chen, 2019](#); [Khatana et al., 2019](#); [Swaminathan et al., 2018](#); [Yan et al., 2020](#)). Without information on these relevant individual-level characteristics, these studies may be underpowered to reliably detect mortality effects of the ACA ([Allen and Sommers, 2019](#)), and may in fact vastly overstate magnitudes ([Black et al., 2019](#)).

¹Authors' calculations using death rates from 2008 to 2013 derived from the publicly-available National Health Interview Survey Linked Mortality File ([National Center for Health Statistics, 2019](#)) for adults ages 55 to 64 with family incomes below 138% FPL and those with family incomes 400% FPL or greater.

In this paper, we provide new evidence of the impact of Medicaid on health by using administrative Medicaid enrollment and mortality data linked to large-scale, individual survey records. We use this novel dataset to examine the impact of the sizeable Medicaid eligibility expansion that occurred in some states under the ACA. In 2014, the ACA expanded eligibility for the Medicaid program to include all adults in families with incomes under 138 percent of the FPL. Previously only pregnant women, adults with disabilities, and very low income parents tended to qualify for Medicaid coverage. Although intended to apply to all states, a 2012 Supreme Court decision made the Medicaid eligibility expansion optional. As a result, only 29 states and the District of Columbia expanded coverage in 2014, with 7 additional states electing to expand over the next several years. Despite non-universal adoption, the ACA Medicaid expansions still represent a historic expansion in insurance coverage. Approximately 13.6 million adults gained Medicaid coverage under the ACA ([Medicaid and CHIP Payment and Access Commission, 2018](#)); for comparison, Medicare enrolled about 19 million elderly beneficiaries after its creation in 1965 ([Bureau of the Census, 1969](#)). We take advantage of variation in state adoption of this large expansion in coverage to compare changes in mortality among individuals in expansion states and non-expansion states.

In contrast to prior research that relies on death certificate data with limited information on individual characteristics, our data include detailed survey measures collected from the 2008 to 2013 years of the American Community Survey (ACS). This large-scale national survey contains approximately 4 million respondents in each year and allows us to observe information on specific characteristics that determine Medicaid eligibility such as income, citizenship status, and the receipt of other social assistance. With this information, we are able to identify individuals targeted by the ACA Medicaid expansions and, in this way, overcome the inherent limitations present in existing studies that rely only on aggregate death records. We focus on those in this group who were between the ages of 55 and 64 in 2014, who are at greater risk of mortality, although we also present results for all non-elderly adults and other age groups. We follow individuals in our sample over time to examine changes in coverage by Medicaid, by linking respondents to administrative records on Medicaid enrollment, and document sizeable changes in annual and cumulative Medicaid coverage in the Medicaid expansion states. Next, we examine whether there is an associated change in mortality by linking the sample to the Census Numident file which contains administrative records on the date of death for all individuals with Social Security Numbers (SSNs) who die in the United States. The Census Numident file allows us to observe mortality rates for our sample through 2017, four years after the initial ACA Medicaid eligibility expansions. We further examine changes in mortality by the underlying cause of death using data from the Mortality Disparities in American Communities (MDAC) project, which links the 2008 wave of the ACS to death certificate records using the National Death Index.

Our analysis shows that the ACA Medicaid expansions reduced mortality among this targeted group. Prior to the expansions, individuals in our sample residing in expansion and non-expansion states had very similar trends in both Medicaid coverage and mortality. At the time of the expansion, the trajectories of these two groups diverged significantly, with expansion state residents seeing increases in Medicaid coverage and decreases in annual mortality rates. In the first year following the coverage expansion, the probability of mortality declined by about 0.089 percentage points, or a 6.4 percent reduction over the sample mean. The estimated impact of the expansions increases over time, suggesting

that exposure to Medicaid results in increasing health improvements. By the fourth year, the expansion reduced annual mortality rates by 0.208 percentage points among expansion state residents. In our supplemental analysis using the MDAC data, we find evidence that healthcare amenable and internal (disease-related) causes of death fell as a result of the expansions, but no evidence of a decline in deaths due to external causes, such as car accidents. We show these estimates are robust to a large number of alternative specifications. We also conduct several placebo tests to assess the validity of our analysis including examining the impact of the expansions on those age 65 or older in 2014 who did not gain Medicaid eligibility; examining the effect on individuals in higher income families who were less likely to be affected; and, conducting the analysis in the pre-ACA period before we would expect to see relative changes in mortality across state groups. We find no relative change in coverage or mortality across expansion and non-expansion states among the elderly or in the pre-ACA period, settings in which no Medicaid expansion occurred. Among those in higher income families, we find small but statistically significant increases in Medicaid coverage as well as small decreases in mortality, consistent with a causal impact of Medicaid on mortality.

Our analysis provides new evidence that expanded Medicaid coverage reduces mortality rates among low-income adults. If we assume that similarly sized mortality reductions would have occurred in the non-expansion states, our estimates suggest that approximately 15,600 deaths could have been averted if the ACA expansions were adopted nationwide as originally intended by the ACA. This highlights an ongoing cost to non-adoption that is relevant to both state policymakers and their constituents.

II Background

Many studies have shown that Medicaid coverage, and the ACA expansion in particular, increases access to and use of health care and reduces financial burden for low-income adults, but evidence on its health effects proves more difficult to document and is less conclusive (Gruber and Sommers, 2019; Soni et al., 2018). Studies that do examine health often rely on self-reported health measures from survey data. The evidence from these studies spans from estimated large or modest improvements in reported health associated with Medicaid expansion (Cawley et al., 2018; Lee and Porell, 2018; Simon et al., 2017; Sommers et al., 2016, 2017), to no effects (Courtemanche et al., 2018a,b; Miller and Wherry, 2019; Sommers et al., 2015; Wherry and Miller, 2016) or even small but marginally significant negative effects (Miller and Wherry, 2017). One concern with self-reported health data is that it may not accurately measure changes in physical health. For example, changes in self-reported health may reflect increasing awareness of health problems or interactions with the health care system, rather than actual changes in physical health. Increased contact with health providers could lead to, e.g., new information about a previously undiagnosed illness and, as a consequence, a worsened self-perception of health. This could bias downwards estimates of the effect of public health insurance on health.² Furthermore, the reliability of self-reported health measures for U.S. adults and their association with objective health measures are documented to be worse among lower socioeconomic status groups (Dowd and Zajacova, 2007, 2010; Zajacova and Dowd, 2011).

²This bias could also operate in the opposite direction if increased interaction with providers improves one's perception of health. See Currie and Gruber (1995) for more discussion.